مسار

برنامج تدريب ميداني

2025

Introduction to Linux Operating System
Part 1

Agenda

- Introduction to Linux OS
- Kali Linux Setup
- Navigating the Linux File System
- Linux Basic Commands
- Linux Files and Permissions
- Linux User Management

Introduction to Linux OS

- Linux is a family of open-source Unix-like operating systems based on the Linux kernel.
- The Linux kernel serves as the core component of the operating system, managing hardware resources and providing essential services to other software on the system.

Linux Distributions:

Unix Operating Systems

- UNIX is a family of multitasking, multi-user operating systems that conform to the Single UNIX Specification (SUS) standard.
- **Key Features:**
  - Time-Sharing System
  - Portable
  - Modular Design
  - Hierarchical File System
  - Examples (All Proprietry UNIX OS's):
    - AIX (IBM)
    - HP-UX (Hewlett-Packard)
    - Solaris (Oracle, formerly Sun Microsystems)

Unix-like Operating Systems (UN*X, *nix or *NIX)

Operating systems that are inspired by or behave like UNIX although not necessarily conforming to or being certified to any version of the Single UNIX Specification (SUS) standard.

Examples:

- Linux.
- BSD Variants (FreeBSD, OpenBSD, and NetBSD).

Kali Linux Setup

VMware Workstation

A virtualization software product that allows users to run multiple operating systems on a single physical machine. It enables the creation and management of virtual machines (VMs), which are isolated software containers that can run different operating systems simultaneously.

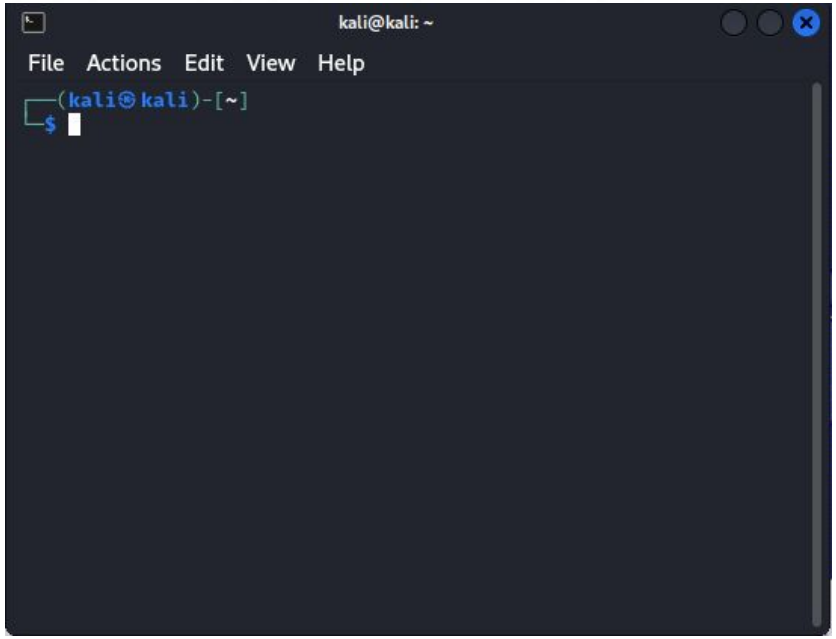
https://www.vmware.com/products/workstation-player/workstation-player-evaluation.html

Kali Linux Download

https://www.kali.org/get-kali/#kali-virtual-machines

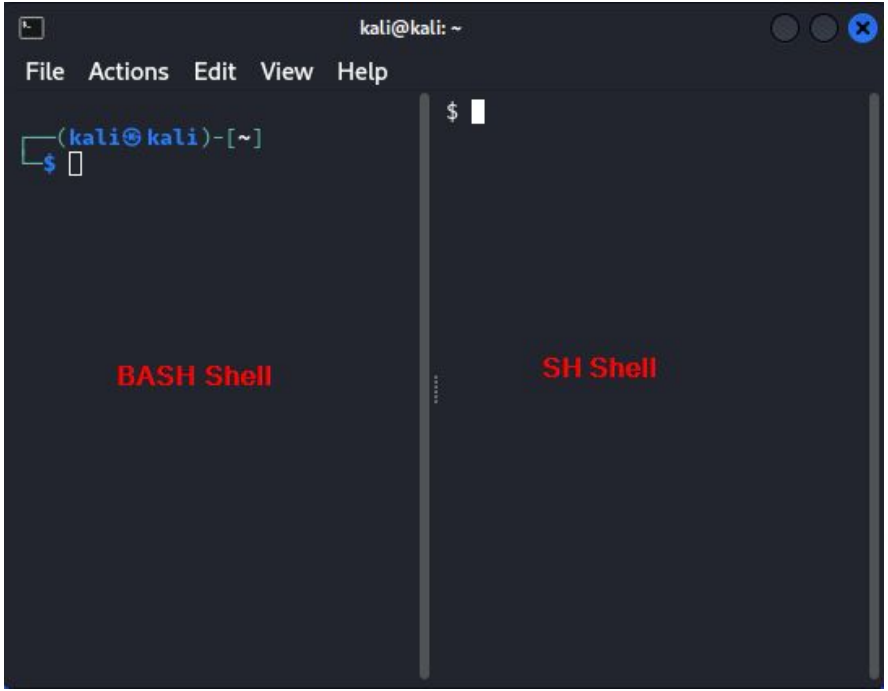
Interacting with Linux OS - Terminal

- A terminal is a program that provides a text-based interface for interacting with the operating system.
- It's often a window or an application in which you can run command-line programs.

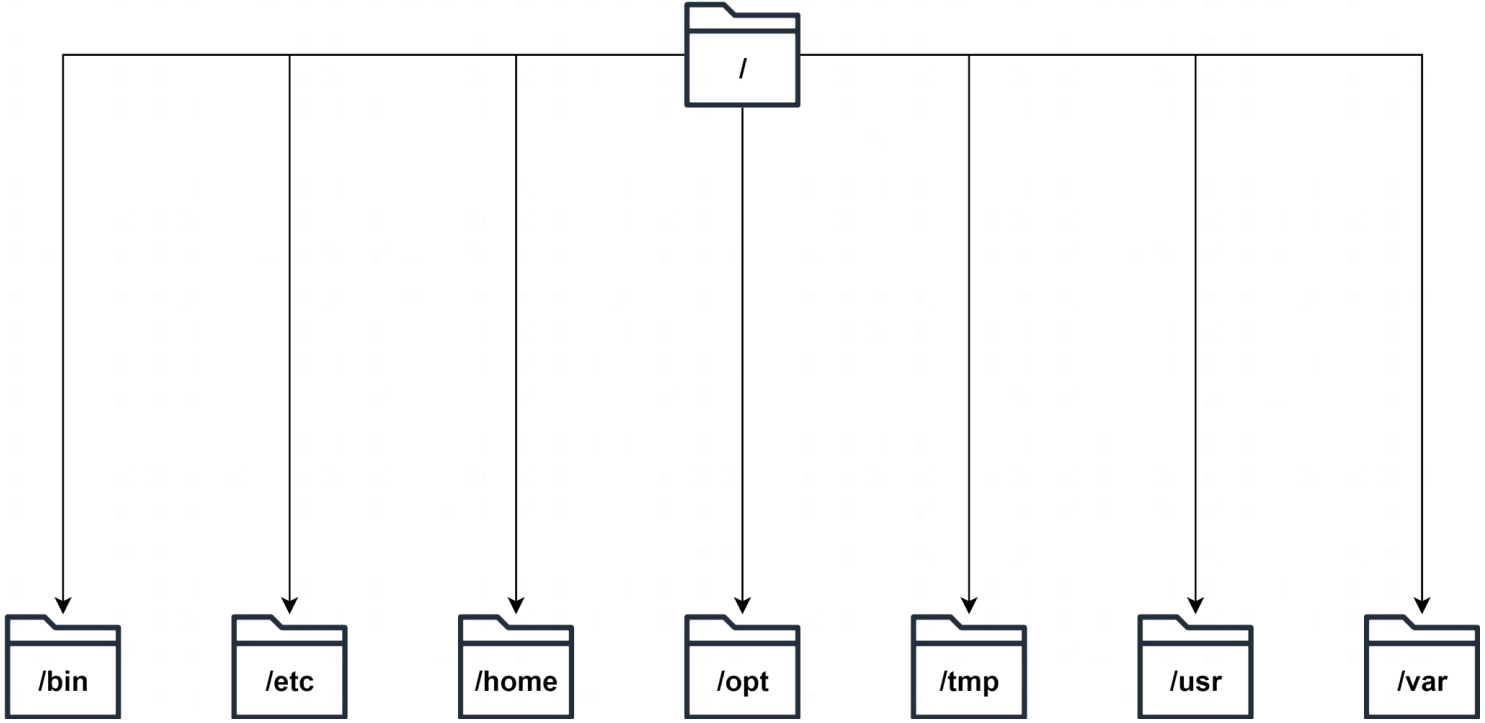


Interacting with Linux OS - $SHELL

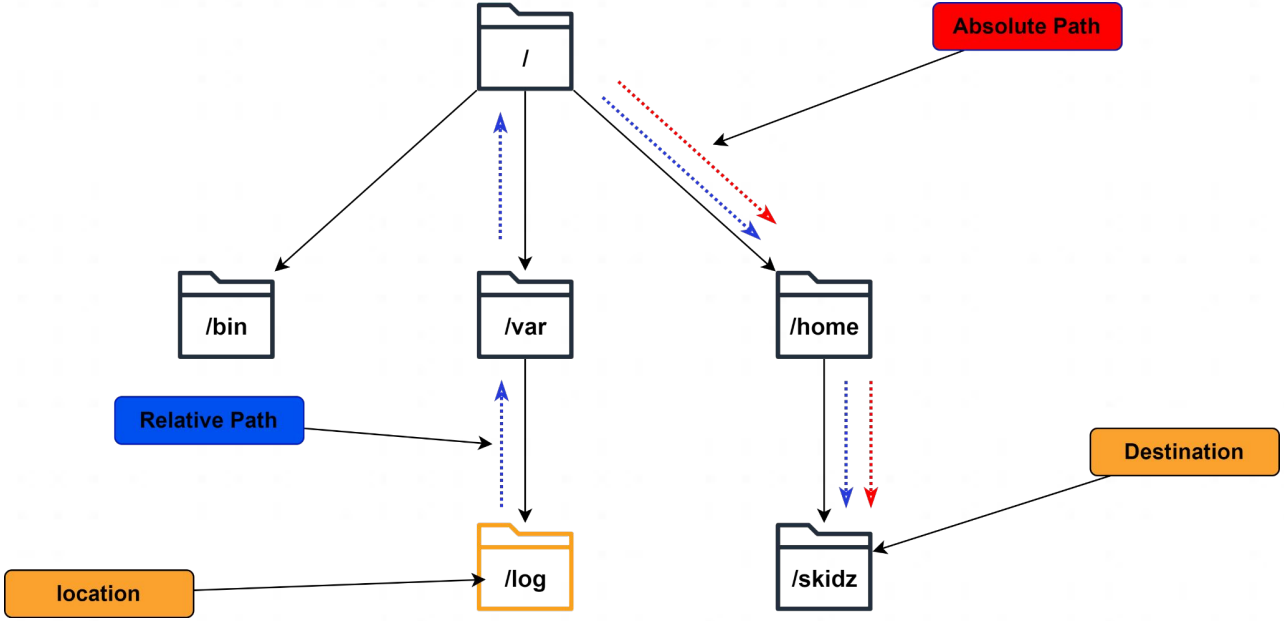
- A **shell** is a command-line interpreter, serving as the bridge between users and the operating system's kernel.
- It processes commands entered by the user, executes them, and displays the results.
- Operating as a layer above the kernel, the shell provides users with indirect access to the kernel's functionalities.



Linux File System



Navigating Linux File System



Linux Basic Commands

- **ls:** Displays the contents of a directory.
  - `ls`
  - `ls -all`
- **cd:** Allows you to change your current working directory.
  - `cd <directory path>`
- **pwd:** Shows the full path of the current working directory.
  - `pwd`
- **cp:** Copies files or directories from one location to another.
  - `cp <file to copy> <destination>`

Linux Basic Commands

- **mv:** Moves or renames files and directories.
  - `mv <file to move or rename> <the new name or destination>`
- **touch:** Creates an empty file or updates the timestamp of an existing file.
  - `touch skidz.txt`
- **mkdir:** Creates a directory.
  - `mkdir <directory name>`

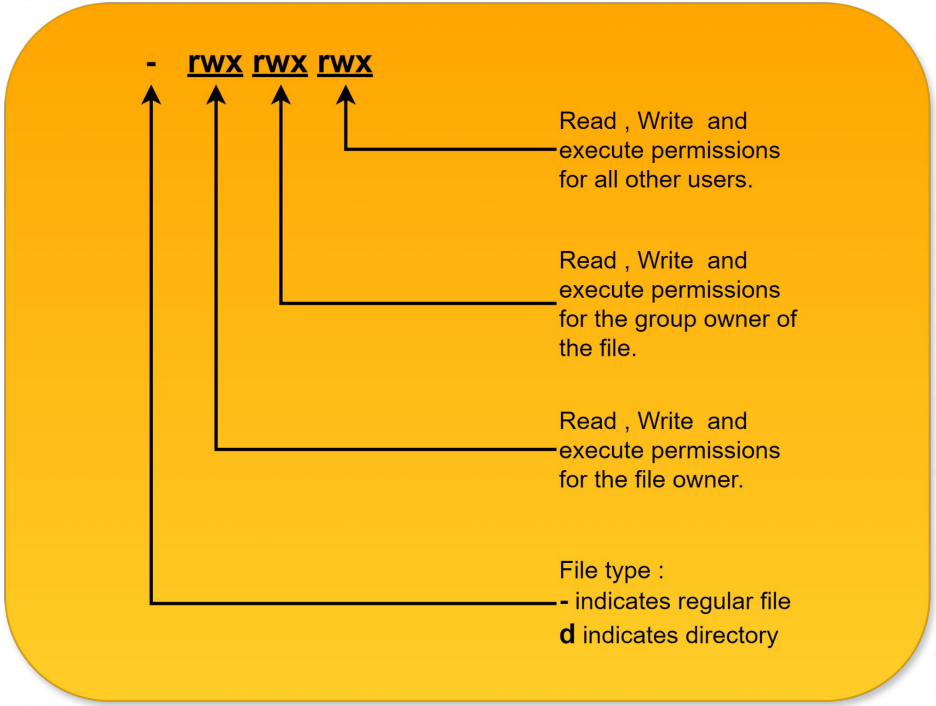
Linux Basic Commands

- **cat:** Displays the contents of a file.
  - `cat <file name>`
- **Pipelines (|):** Allows the output of one command to serve as the input for another.
- **grep:** Searches for a pattern in files and displays the lines that match.
  - `cat skidz.txt | grep “secret”`

Linux Basic Commands

- **Standard input, output, and error:**
  - STDIN '<': is used to take input from a file.
    - `cat < skidz.txt`
  - STDOUT '>': is used to redirect output to a file.
    - `cat skidz.txt > skidz.txt`
  - STERR '2>': is used to represent error messages or diagnostic outputs from a program.
    - `ls non_existent_file 2> error.txt`
- **echo:** display/output text to the terminal
  - `echo "secret" > skidz.txt`

Linux Files and Permissions

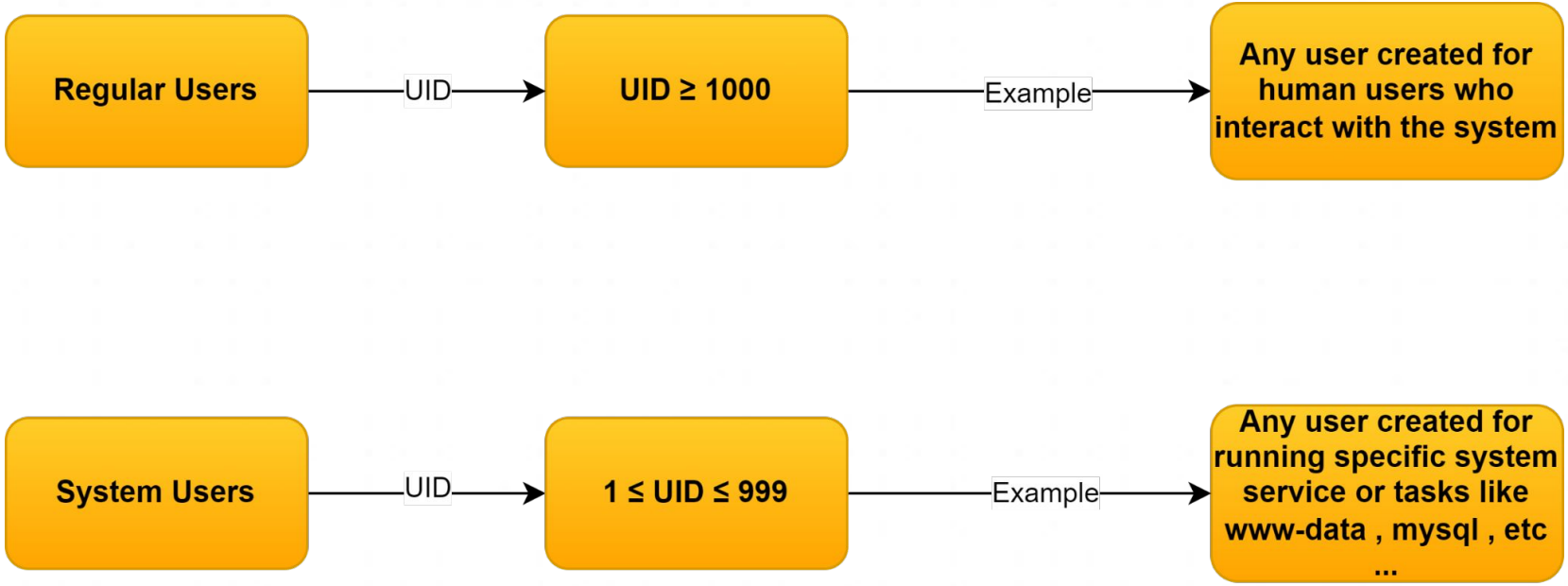


Linux Files and Permissions

Advance permissions:

- **SUID:** When the SUID permission is set on an executable file, the program is executed with the privileges of the file owner, rather than the user who is executing the program.
  - `chmod u+s <executable>`
- **SGID:** If set on a file, it allows the file to be executed as the group that owns the file.
  - `chmod g+s <executable>`
- **Sticky bit:** This permission does not affect individual files. However, at the directory level, it restricts file deletion. Only the owner of the file, the root (superuser), and the directory's owner can remove (or rename) the file within that directory. (e.g. /tmp)
  - `chmod +t <directory>`

Linux User Management



Linux User Management - /etc/passwd

- **/etc/passwd:** contains basic information about user accounts on the system using the following format:

```
root:x:0:0:root:/root:/bin/bash
ubuntu:x:1000:1000:Ubuntu,,,:/home/ubuntu:/bin/bash
sshd:x:128:65534::/run/sshd:/usr/sbin/nologin
phantom:x:1001:1001::/home/phantom:/bin/sh
mysql:x:129:137:MySQL Server,,,:/nonexistent:/bin/false
skidz:x:1002:1002::/home/skidz:/bin/sh
```

Linux User Management - /etc/shadow

- **/etc/shadow:** stores the secure information related to user accounts, particularly the hashed passwords and other security-related data.

```
root:$y$j9T$J2LkYLd.CafOrKLWyY2lS/$2Mz10mxKdnX0j6V94P0wxY9oGZkgy634Hb/2wpl6Ci.:19699:0:99999:7:::
ubuntu:$y$j9T$blJyqHv0AKwCZQk3w3iwg0$/Rd0h5UGpCUFuGiz5oGT5wCvONZNfC/rxvHTmHKdhw9:19104:0:99999:7:::
sshd:*:19104:0:99999:7:::
phantom:$1$1y6fnPgi$tY4RI3ZtP5qni0KPruTit.:19699:0:99999:7:::
mysql:!:19699:0:99999:7:::
skidz:$y$j9T$ILC66kSmpiQYgv6WH6M/y1$ElmIPquvTCEUTnV6/PPEWk/q7CgNxiLGmW3gUgXX3tD:19890:0:99999:7:::
```

Linux User Management - /etc/group

- **/etc/group:** a system file on Unix-like operating systems, including Linux, that stores information about user groups.

```
bin:x:2:
sys:x:3:
adm:x:4:kali
users:x:100:kali
mysql:x:122:
sslh:x:127:
kali:x:1000:
admins:x:1002:skidz,kali
```

Linux User Management - Commands

- Create new user
  - `useradd <username>`
  - `useradd -m -s /bin/bash  <username>`
- Change user password
  - `passwd <username>`
- Delete user
  - `userdel -r <username>`